Learning Outcome:

After completing this exercise, you shall be able to:
• Load and explore a real-world dataset using Pandas by examining its structure, variables, statistics, and data distributions.
• Clean and prepare data for machine learning by handling missing values, identifying relevant features, and preparing the target and independent variables.
• Build, evaluate, and analyze an SVC classification model using appropriate performance metrics and visualizations to understand passenger survival patterns.

Problem Statement:

In this hands-on exercise, you are required to work with the Titanic dataset to perform data exploration, cleaning, feature preparation and visualization using Python. You will load the dataset into a
DataFrame, explore its numerical and categorical variables, handle missing values, identify the target and
independent features, and build an SVC classification model to predict passenger survival. You will then
evaluate the model using accuracy, confusion matrix, precision, recall, and F1-score, and create visualizations to analyze survival patterns based on gender, passenger class, and fare. The objective is to
understand how a real-world dataset can be explored, prepared, analyzed, and used to build and evaluate a
machine learning classification model.
You can use Jupyter Notebook, Google Colab, VS Code, or any other Python programming environment to write and execute your code.

Dataset Description:

The Titanic dataset contains information about passengers who travelled on the RMS Titanic, which sank on April 15, 1912, after colliding with an iceberg. The dataset includes details such as passenger
class, gender, age, number of siblings/spouses, number of parents/children, ticket fare, and port of embarkation. It also contains the Survived variable, which indicates whether a passenger survived the disaster (1) or did not survive (0). In this exercise, the dataset will be used to explore the factors related to
passenger survival and to build a Support Vector Classifier (SVC) that predicts whether a passenger survived.
Use the titanic cvs dataset in the repo